

WASP-43b

R-0649

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_170310 · created 2026-07-25T21:40:27+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457822.8647 +/- 0.0013 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.168 +/- 0.057
Transit depth (Rp/Rs)^2	2.82 +/- 1.92 [%]
Orbital Inclination (inc)	86.99 +/- 2.97
Airmass coefficient 1 (a1)	0.96 +/- 0.48
Airmass coefficient 2 (a2)	0.035 +/- 0.078
Scatter in the residuals of the lightcurve fit is	50.65 %
Best Comparison Star	None
Optimal Aperture	4.01
Optimal Annulus	5.83
Transit Duration (day)	0.0594 +/- 0.0051

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.168 ± 0.057 vs 0.1594 (deviation 0.15σ)
Duration fit vs published	0.0594 d vs 0.0512 d (deviation 1.6σ)
Mid-time O–C	-1.3 min
Depth SNR	2.9
Residual scatter	50.65 %

Light curve

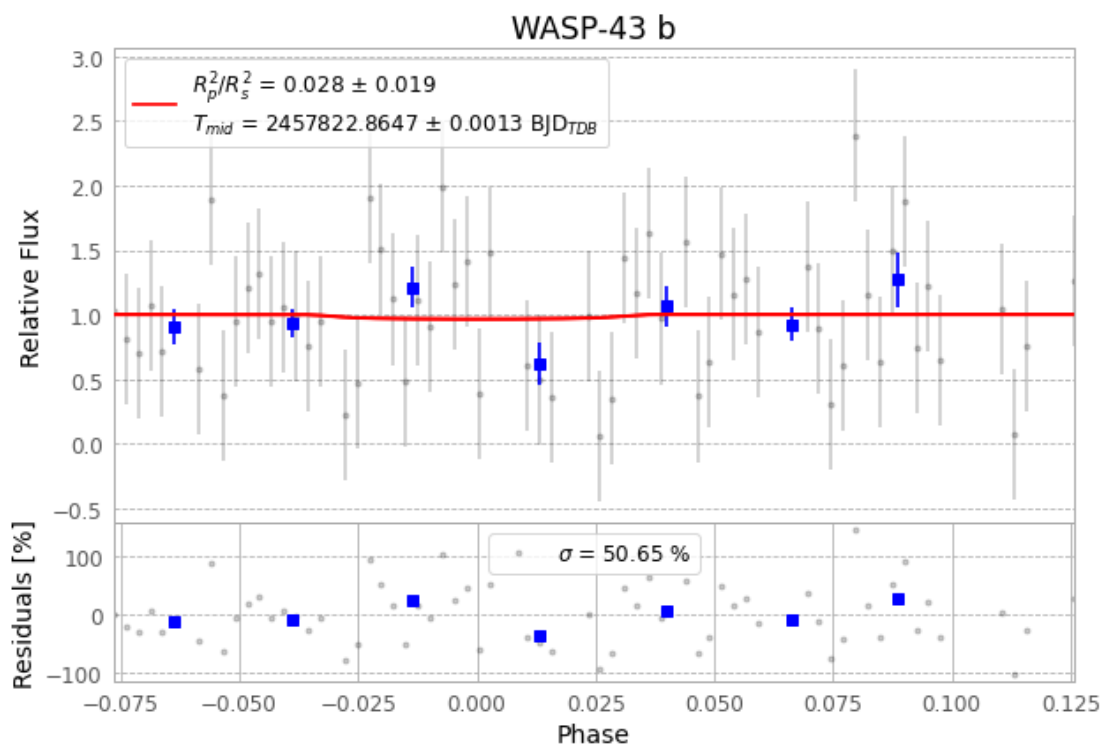


Figure 1 — FinalLightCurve_WASP-43 b_2017-03-10.png (SHA-256 2739a117129af632...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [294, 196], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 1ee12c8afad5b135...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494

Data provenance

80 FITS frames (53 MB), WASP-43170310070612.FITS → WASP-43170310110912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-170310.log (fa6323de4b74...), FinalLightCurve_WASP-43 b_2017-03-10.png (2739a117129a...), FinalLightCurve_WASP-43 b_2017-03-10.csv (c287eae87be...)

Compute resources

Wall time 140.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-25 21:40Z.